ANA BULLETIN NO. 306m

27 DECEMBER 1963

Superseding
ANA Bulletin No. 306k
16 January 1956

AIR FORCE-NAVY AERONAUTICAL BULLETIN

ENGINES, AIRCRAFT TURBINE AND JET, DESIGNATION OF

1. Contents.- This bulletin contains the list of letters and numerals, and describes the method of their assignment and arrangement used by the Air Force and by the Navy in establishing the type and model designation of aircraft turbine and jet engines.

2. New designations.- A new engine designation will be assigned when, in the judgement of the assigning Service, any one of the following conditions is met:

(a) A need (either administrative or technical) exists for reference to, or definition of, a particular engine configuration or design which has not previously been assigned a designation. Normally, this will occur as a result of new engine development programs, or upon the initial entry into the Services inventory of "off the shelf" engines, and will involve a newly assigned type indicator.

(b) An engine configuration or design with an existing designation is so revised that there is a significant change in performance, flight safety, installation, or the interchangeability of the complete engine in an aircraft is affected, and the change is not made immediately retroactive to all engines of the same designation previously delivered. Ordinarily, this will involve the assignment of a new model number only. However, when the design change alters the engine fundamentally, a new type indicator may be assigned.

(c) Design changes are made to an engine, with an existing designation, to the extent that a substantial number of parts or sub-assemblies (or both) are no longer interchangeable with the corresponding parts or sub-assemblies furnished in accordance with the approved parts list for the original model engine, but the changes do not affect the performance, installation, or interchangeability of the complete engine in an aircraft. Ordinarily, this will involve only the addition or change of the suffix letter in the model indicator, although change of the model number may be indicated if the quantity of parts affected is sufficiently large.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:03Z
Check the source to verify that this is the current version before use.

3. Special designations.- The letters "X" and "Y" may be used at the discretion of the Services for the purpose of signifying development engines, as applicable. Generally, the prefix letter "X" shall be used to denote an engine under development which has not reached the stage of successfully completing the official preliminary flight rating tests, and the prefix "Y" shall be used to denote an engine still in the development category, but which has been cleared for limited flight use by successfully completing the preliminary flight rating tests. When used, such letters shall precede the designation arrangement described herein.

4. Designation.- The complete engine designation will consist of three parts. These are the type indicator, manufacturer's symbol, and the model indicator. A given engine design and configuration will have only one designation for both Services. For example, should the Navy desire to use an engine bearing Air Force type and model numbers, the Navy will use those numbers without change for all designation purposes. Further, should the Air Force desire to use a Navy approved type engine, but require changes to the Navy model of that type, the Air Force will use the Navy type designation and assign its own model indicator to the modified engine.

4.1 Type indicator.- The type indicator part of the engine designation will consist of the appropriate type letter (or letters) symbol selected from 4.1.1, together with a type number assigned in accordance with 4.1.2.

4.1.1 Type letter symbols.- Type letter symbols shall conform to:

Letter symbols	Engine type
J	Turbojet
T	Turboshaft or Turboprop
TF	Turbofan
RJ	Ram jet

4.1.2 Type numerals.- The type numerals used in connection with the type letter (or letters) will be assigned progressively by the Services, and shall begin with the number 30 for the Navy, and the number 31 for the Air Force. Even numbers will be assigned by the Bureau of Naval Weapons to types approved by the Navy, and odd numbers will be assigned by the Systems Engineering Group, Research and Technology Division, Wright-Patterson Air Force Base, to types approved by the Air Force. The type numbers are arbitrary, and do not represent any characteristics of the units involved.

4.2 Manufacturer's symbols.- The second part of the designation will consist of a dash and a letter(s) symbol indicating the manufacturer, as follows:

MANUFACTURER'S NAME	LETTER SYMBOL
ACTIVE	
AiResearch Division, Garrett Corporation	G
Allison Division, General Motors Corporation	A
Boeing Company	BO
Continental Aviation and Engineering Corporation	T
General Electric Company	GE
Lycoming Division, Avco Corporation	L

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:03Z
Check the source to verify that this is the current version before use.

MANUFACTURER'S NAME (Continued)	LETTER SYMBOL
ACTIVE	
Marquardt Corporation	MA
Orenda Engines - A Division of Hawker Siddeley Canada, Ltd.	OEL
Pratt and Whitney Aircraft Division, United Aircraft Corporation	P
Rolls Royce, Ltd.	RR
Solar Aircraft Company	S
United Aircraft of Canada, Ltd.	CP
Wright Aeronautical Division, Curtiss-Wright Corporation	W
INACTIVE	
Aerojet-General Corporation	AJ
Aeromarine Company	AM
American Helicopter Company, Inc.	AH
Buick Motor Division, General Motors Corporation	B
Chevrolet Motor Company, Division General Motors Corporation	C
Chrysler Corporation	D
Fairchild Engine Div., Fairchild Engineering and Airplane Corporation	R
Ford Motor Company	F
Frederick Flader Company	FF
G. M. Giannini and Company	GN
Hiller Helicopters	H
McDonnell Aircraft Corporation	MD
Northrop Company	NH
Packard Motor Car Company	PM
Studebaker Motor Company	ST
Westinghouse Electric Corporation	WE

4.3 Model indicator.- The third part of the designation, the model indicator, will consist of a dash and a model number, or a dash and a model number with a suffix letter.

4.3.1 Model number.- The model number will be assigned for a specific configuration of a given type engine. Air Force model numbers for each type of engine will begin with -1 and will continue with consecutive odd numbers. Navy model numbers for each type of engine will begin with -2 and will continue with consecutive even numbers. All even model numbers will be assigned by the Bureau of Naval Weapons, including those applied to Air Force approved engine types. All odd numbers will be assigned by the Systems Engineering Group, Research and Technology Division, Wright-Patterson Air Force Base, including those applied to Navy approved engine types.

4.3.2 Suffix letter(s).- Suffix letters will be assigned consecutively, starting with the letters A, B, C, etc., except for the letters I, O, and M. In addition, the letter "W" is reserved for use with the basic model number with suffix letter, to indicate engines with water injection or other performance augmentation features. When so used, the letter "W" will precede any other suffix letter which is assigned in the model indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:03Z
Check the source to verify that this is the current version before use.

5. Examples.- The following hypothetical examples illustrate the arrangement and significance of the subject designations:

J30-A-2	First Navy model of first Navy turbojet type. (Made by Allison)
J31-W-1	First Air Force model of first Air Force turbojet type. (Made by Wright)
J31-GE-1	First Air Force model of first Air Force turbojet type. (Wright engine made by General Electric)
J35-GE-2	First Navy model of third Air Force Turbojet type. (Made by General Electric)
J35-GE-3B	Second Air Force model of third Air Force turbojet type with second design change(s) to the basic model, not affecting performance, flight safety, installation, or interchangeability. (Made by General Electric)
T34-P-3	Second Air Force model of third Navy turboprop type. (Made by Pratt and Whitney)
RJ35-T-6	Third Navy model of third Air Force ram jet type. (Made by Continental)
PJ36-MA-7	Fourth Air Force model of fourth Navy pulse jet type. (Made by Marquardt Corporation)
XJ34-WE-2	First Navy model of third Navy turbojet type, in development status prior to preliminary flight rating test. (Made by Westinghouse)
YT35-W-2	First Navy model of third Air Force turboprop type, in development status cleared for flight test. (Made by Wright)
J37-BO-5W	Third Air Force model (water injection) of fourth Air Force turbojet type. (Made by Boeing Company)
TF33-P-7	Fourth Air Force model of second Air Force turbofan type. (Made by Pratt and Whitney)
J69-A-9WC	Fifth Air Force model (water injection) of the twentieth Air Force turbojet type, with third design change to the basic model, not affecting performance, flight safety, installation, or interchangeability. (Made by Allison)

6. Manufacturer's code symbols.- These letter symbols are not to be confused with those codes established by the Federal Supply Code for Manufacturers, H4, which covers code identification symbols of principal manufacturers of equipment and parts.

-ooo-

Source: https://assist.dla.mil -- Downloaded: 202610-26T14:03Z
Check the source to verify that this is the current version before use.